A reaction that takes energy in, usually as

The chemicals that react together

products

fuel

a fuel is burnt in plenty of air

In chemical equation, reactants come ...

... after the arrow

No new substance is made, but there is a change of

reactants

To break up with heat

conserved

carbon + oxygen ⟶ carbon dioxide

Chemicals combine to make a new

Three states of matter

physical change

Equation to show the reaction between carbon and oxygen

hydrogen and carbon

heat and light

oxygen

Another word for ‘burning’

Three parts of the fire triangle

Hydrocarbons always contain …

Substances formed from a chemical reaction

In a chemical equation the products come …

Product of incomplete combustion

Stores energy in chemical store which can be released as

Forces that hold atoms together

chemical reaction

before the arrow

Equation to show magnesium burning in air

thermal decomposition

Complete combustion happens when…

oxygen, heat and fuel

If carbon dioxide is present, lime water will go

heat

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When the amount of something does not change

combustion

A chip pan fire is usually stopped by putting a damp cloth over it as it starves it of ...

magnesium + oxygen → magnesium oxide

Product when sulfur burns in oxygen

carbon monoxide

endothermic reaction

Solid, liquid, gas

sulfur dioxide

A reaction in which energy is given out, usually as heat or light

chemical bonds

cloudy/ milky

exothermic reaction

During combustion energy is transferred to the surroundings as

...

Teaching notes and answers

Ask students to solve the puzzle by matching up the correct edges to form a 4x4 square.

fuel	Stores energy in a chemical store which it can release as heat
chemical reaction	Chemicals combine to make a new substance
physical change	No new substance is made, but there is a change of state
reactants	The chemicals that react together
products	Substances formed from a chemical reaction
In a chemical equation the products come …	after the arrow
In a chemical equation reactants come …	before the arrow
thermal decomposition	To break up with heat
conserved	When the amount of something does not change after a process takes place
combustion	Another word for ‘burning’
Three parts of the fire triangle	oxygen, heat and fuel
Complete combustion happens when …	A fuel is burnt in plenty of air
Forces that hold atoms together in molecules	chemical bonds
During combustion energy is transferred to the surroundings as …	heat and light
Equation to show the reaction between carbon and oxygen	carbon + oxygen ⟶ carbon dioxide
Equation to show magnesium burning in air	magnesium + oxygen ⟶ magnesium oxide
Product when sulfur burns in oxygen	sulfur dioxide
If carbon dioxide is present, lime water will go …	cloudy/milky
exothermic reaction	A reaction in which energy is given out, usually as heat or light
endothermic reaction	A reaction that takes energy in, usually as heat
Product of incomplete combustion	carbon monoxide
Three states of matter	solid, liquid and gas
A chip pan fire is usually stopped by putting a damp cloth over it as it starves it of …	oxygen
Hydrocarbons always contain …	hydrogen and carbon